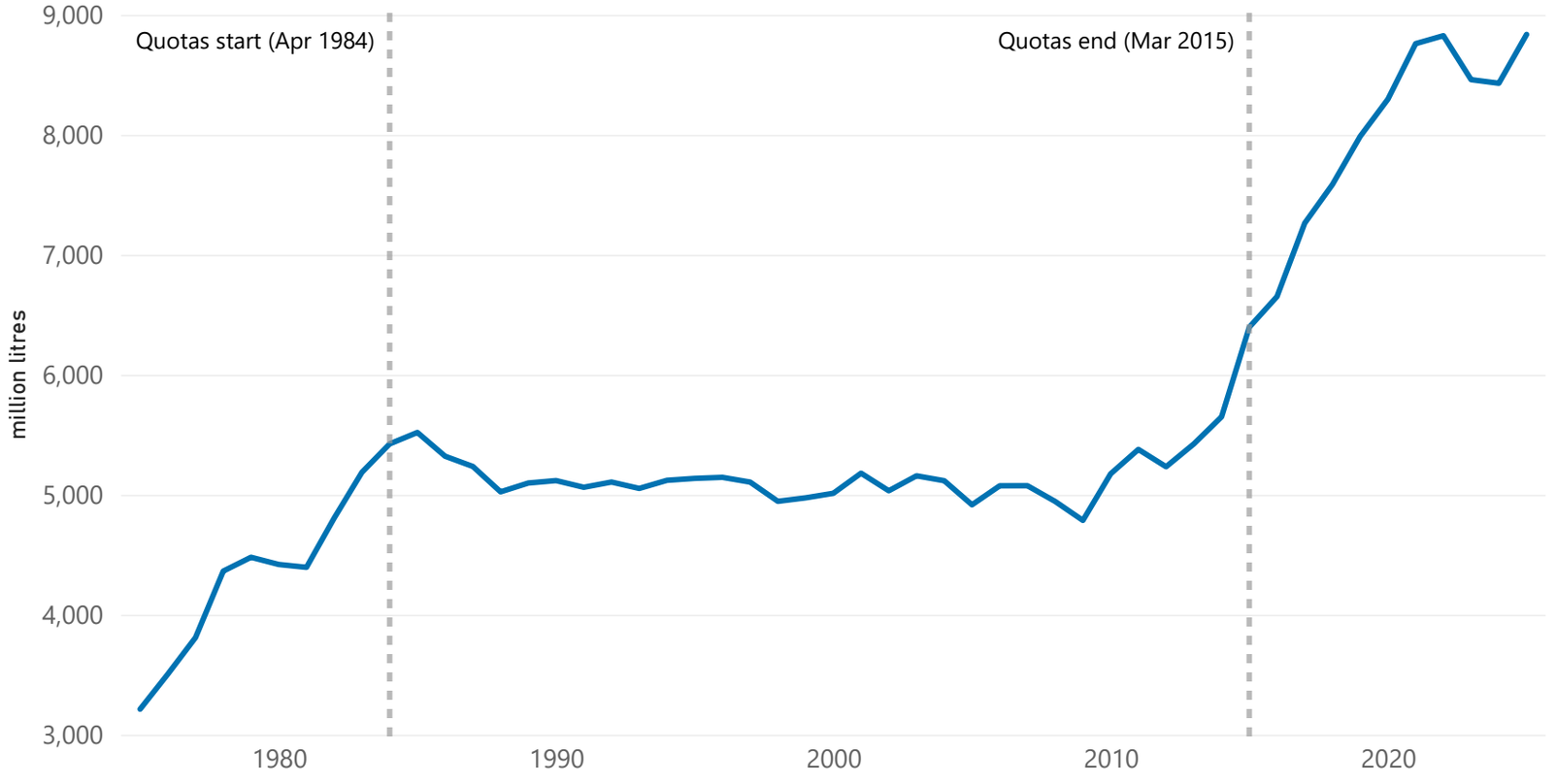
Irish milk intake by year, complete years only (million litres, CSO)



Milk quotas limited how much milk Irish farms could produce from April 1984 to March 2015 (dashed lines). Yearly intake was almost flat under quotas and has grown since they ended.

Source: CSO table AKM01, milk intake by creameries and pasteurisers. Only years with all 12 months are shown.

Milk intake that year (million litres)

8,835

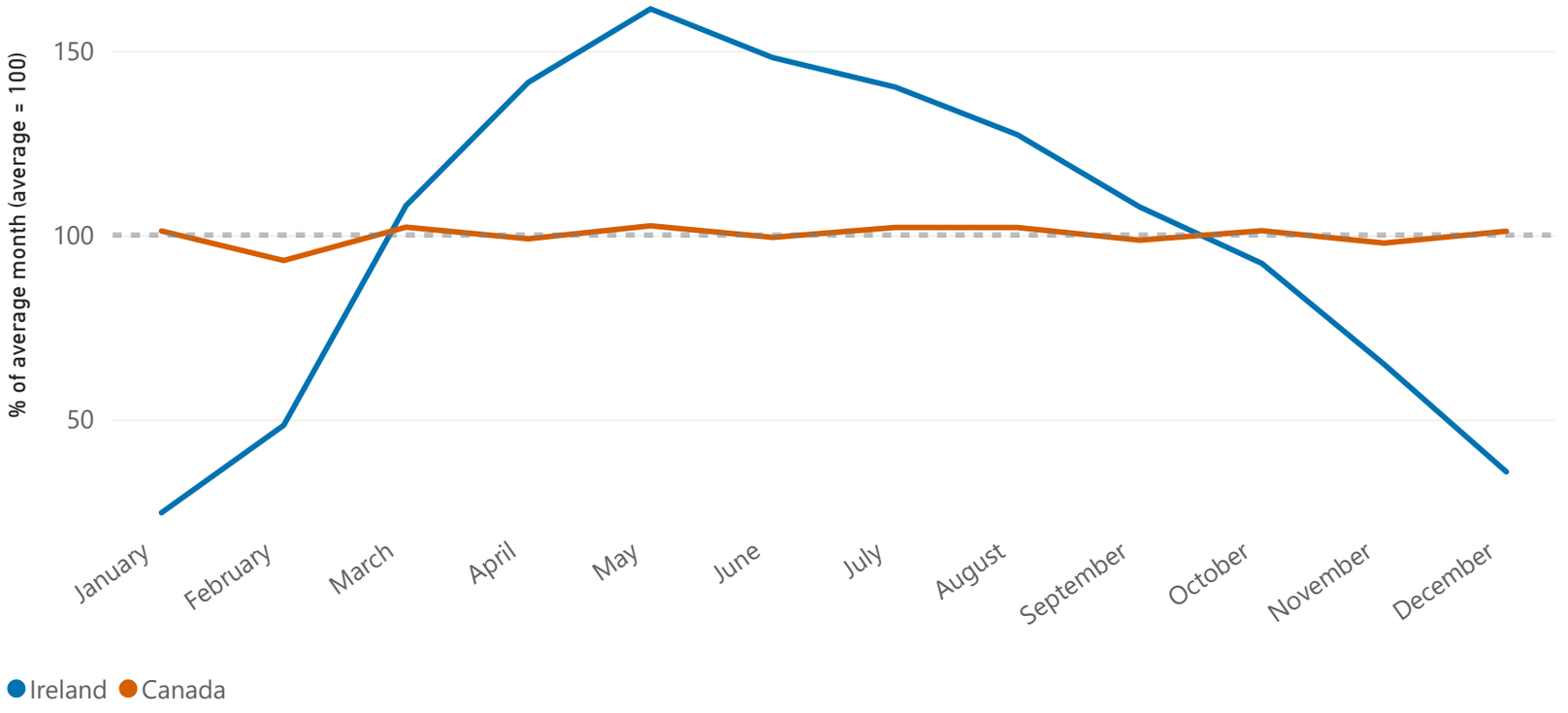
Growth since 2014

+56.4%

Latest complete year

2025

Monthly milk supply as % of each country's average month, selected years (CSO, Statistics Canada)



year

2016 2025

Each month is shown as a % of the country's own average month (average = 100), so the chart compares the shape of the year, not the size of the two dairy sectors.

With the default years in the slicer, Irish supply peaks in May and is lowest in January. Canadian supply stays close to 100 all year. Plants sized for the peak month would use about 62% of their capacity in an average month in Ireland and about 98% in Canada.

Move the year slider to an earlier period to see how the Irish season has changed.

Sources: CSO table AKM01 (Ireland), Statistics Canada table 32-10-0113-01 (Canada).

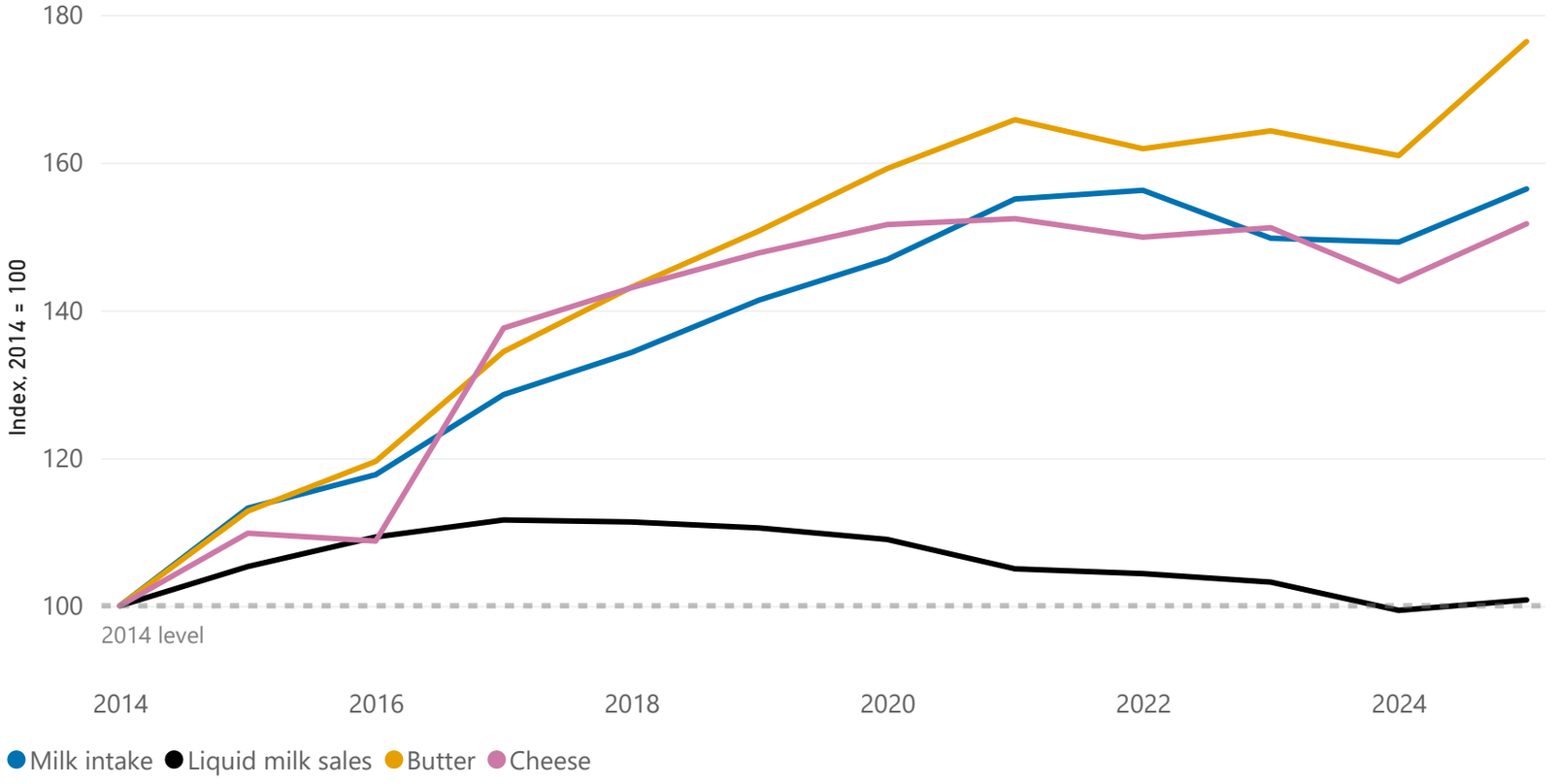
Ireland: average month as % of peak month

62%

Canada: average month as % of peak month

98%

Milk intake, liquid milk sales, butter and cheese, index 2014 = 100, complete years only (CSO)



Where did the extra milk go?

Since 2014, the last full quota year, milk intake grew by +56.4%.

Liquid milk sales (drinking milk) hardly changed: +0.8%.

Butter grew faster than intake (+76.4%), and cheese grew about as fast (+51.7%).

So the extra milk went into processed products such as butter and cheese, not into drinking milk.

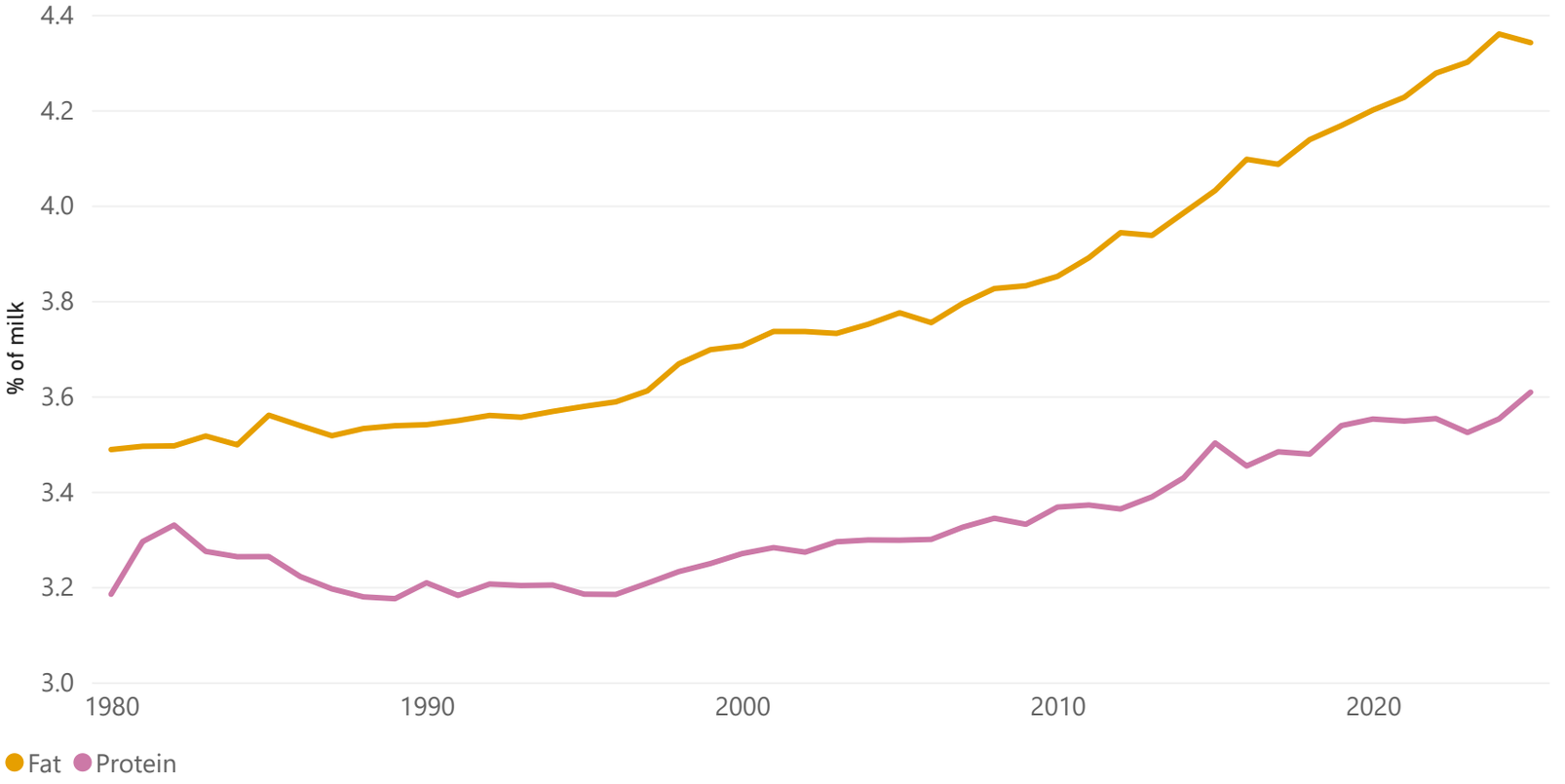
How to read the chart: each line is an index, 2014 = 100. A value of 150 means 50% more than in 2014.

Complete years only. Skimmed milk powder is not shown because of missing months. CSO cheese data ends in December 2025.

Sources: CSO AKM01 (milk intake), AKM02 (liquid milk sales), AKM03 (butter, cheese).

Milk intake vs 2014	Liquid milk vs 2014	Butter vs 2014	Cheese vs 2014
+56.4%	+0.8%	+76.4%	+51.7%

Fat and protein content of milk intake by year, weighted by intake, complete years only (CSO)



Milk is getting richer

Fat content is now 4.34% and protein 3.61%.

Since 2014, fat rose by +0.36 points and protein by +0.18 points. Fat rose about twice as fast.

The chart shows the same long-term pattern since 1980: both rising, fat faster than protein.

More fat and protein per litre means more butter and cheese from the same milk.

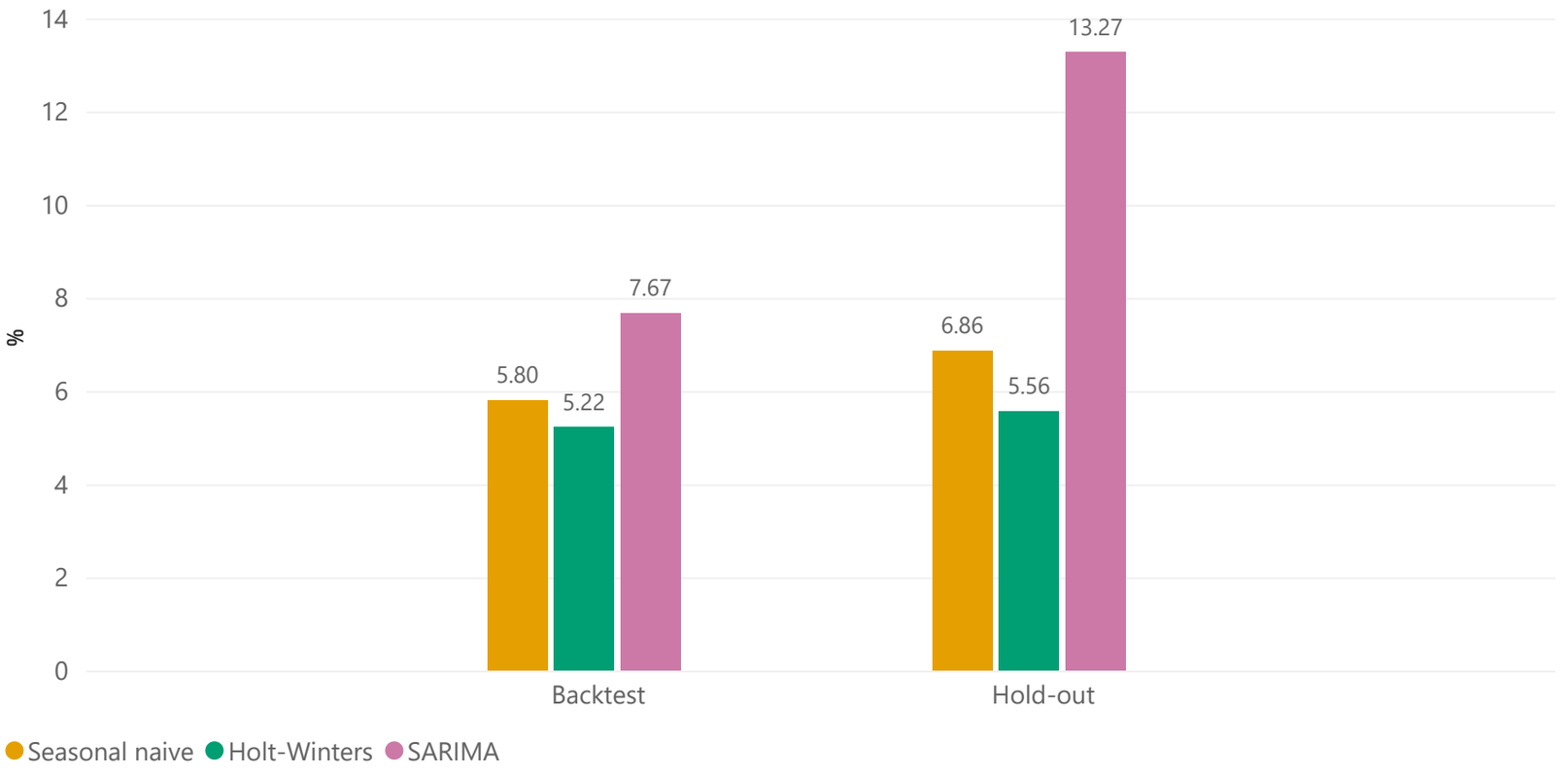
How the yearly values are calculated: each month is weighted by its milk intake, so May, with the most milk, counts most. A simple average of the 12 months would give low-volume winter months too much weight.

Complete years only.

Source: CSO AKM01 (milk intake, fat and protein content).

Fat, latest year	Protein, latest year	Fat vs 2014	Protein vs 2014
4.34%	3.61%	+0.36 points	+0.18 points

Average forecast error (MAPE, %) of 12-month forecasts by model, Irish milk intake (CSO)



Test	Backtest						Hold-out			
Model	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Seasonal naive	8.35	7.85	7.19	4.08	4.77	2.55	8.12	9.86	5.60	3.85
Holt-Winters	8.27	7.78	3.46	4.55	4.87	2.37	8.93	5.98	4.16	3.17
SARIMA	5.02	9.30	23.91	1.76	3.80	2.21	16.79	21.63	10.59	4.05

How accurate is each forecast?

Each model forecasts the next 12 months of milk intake, retrained every December on all earlier data. Error is measured as MAPE: the average gap between forecast and actual, as a % of actual. Lower is better.

Backtest: years of steady growth.
Hold-out: years after growth stopped.

Findings

- Holt-Winters has the lowest average error in both tests: 5.22% and 5.56%.
- The seasonal naive forecast (each month the same as last year) is second: 5.80% and 6.86%. A more complex model has to beat this baseline to be worth using.
- SARIMA has the highest average error: 7.67% and 13.27%. In some years it is the most accurate model (1.76% in 2020), but a few large misses (23.91% in 2019, 21.63% in 2024) pull its average up.

For planning, Holt-Winters gives a stable monthly profile, but next year's level is better treated as a range of scenarios than as one number.

The last year in the table is a partial year.